

MEC Defense: Securing Smart Stadium Critical Infrastructure

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Goals

- Develop an open source software based security framework deployable on MEC networks
- Adopted solutions may not degrade guest traffic when system network is under attack
- Real-time problem detection with automatic alert and mitigation, if possible
- Present filtered logs through a dashboard for system administrators
- Redirect and contain threats for forensic analysis



Dashboard

App Catalog 2

Name	Provider	Version
Test Server MEC App	IT	1.0
Video Processin Tool Test	IT	1.0

App Instances 2

Expand	Name ↑	Domain/Node ↑	CPU (%) ↑	Mem (%) ↑	Open
▼	Test App	-	-	-	
▼	Demo with agent	-	-	-	

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MEC network based on the architecture and framework developed in the OSM-MEC project by Pedro Ferreira, Pedro Escaleira, Diogo Gomes, and Rui Aguiar, at Instituto de Telecomunicações and University of Aveiro, Portugal, enabling the integration of Multi-access Edge Computing with NFV orchestration through ETSI Open Source MANO (OSM) for automated deployment and management of edge services.



Personas



- João Malhoa
- Network Engineer
- Leases a MEC system for big events (e.g football games)
- Wants a quick framework implementation to assure security within the network



- Chuck Keith
- SOC Analyst
- Works in a SOC group at a private company
- Wants to monitor real-time packet traffic in the MEC system to prevent and assure security to Apps and Services present within the system



User Story

manager.name: mec-siem agent.id: 006  rule.groups: suricata X

	↓ timestamp	agent.name	rule.description
	Mar 16, 2026 @ 21:38:26.4...	video-agent	Suricata: Alert - Simulated Network Attack Detected

- A SOC Analyst
- Wants to see logs from MEC Apps in the system
- To interpret the logs and react to alerts

Table	JSON
t _index	wazuh-alerts-4.x-2026.03.16
t agent.id	006
t agent.ip	10.42.0.34
t agent.name	video-agent
t data.alert.category	Attempted Information Leak
t data.alert.severity	1
t data.alert.signature	Simulated Network Attack Detected
t data.dest_ip	8.8.8.8
t data.event_type	alert
t data.src_ip	192.168.1.50
data.timestamp	Mar 16, 2026 @ 21:40:00.000
t decoder.name	json
t id	1773697106.2264782
t input.type	log
t location	/var/log/app-logs/child.log
t manager.name	mec-siem
t rule.description	Suricata: Alert - Simulated Network Attack Detected
# rule.firedtimes	1
t rule.groups	ids, suricata
t rule.id	86601
# rule.level	3
rule.mail	false
timestamp	Mar 16, 2026 @ 21:38:26.486



Requirements

- Functional SOC with real time visualisation
- Custom rules and filters
- Identify threats inside the network
- Isolate each app
- Log correlation
- Traffic segmentation
- Automatic response systems



Non-functional requirements

Low latency

Scalability

Reliability

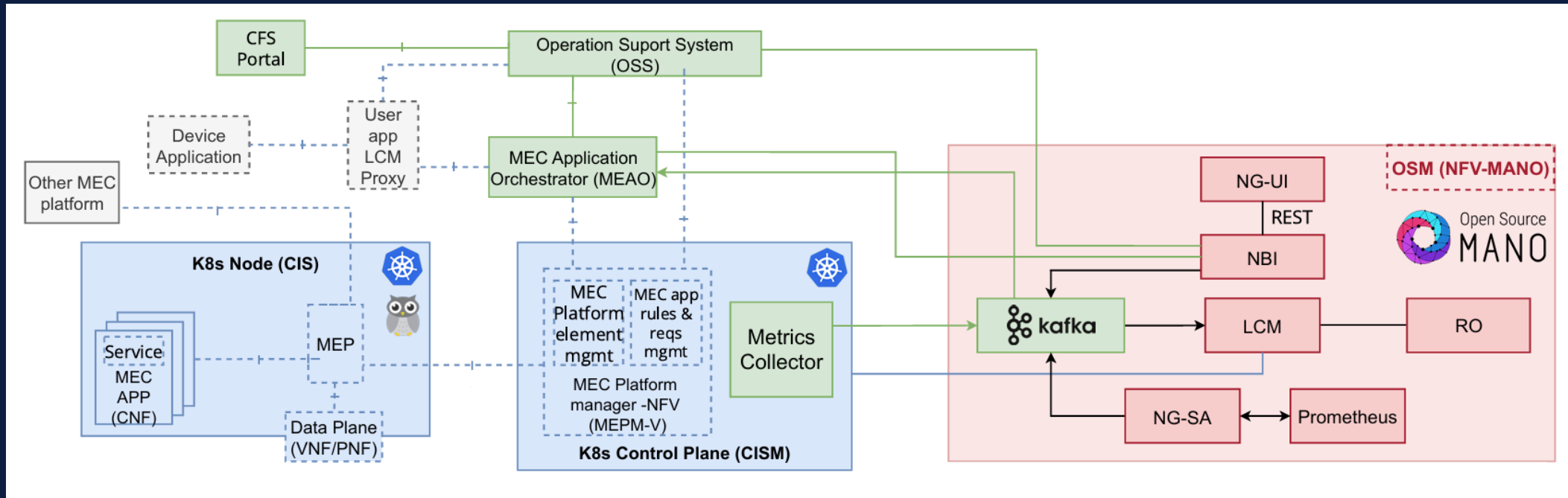
High
Throughput

Auditability

Resilience



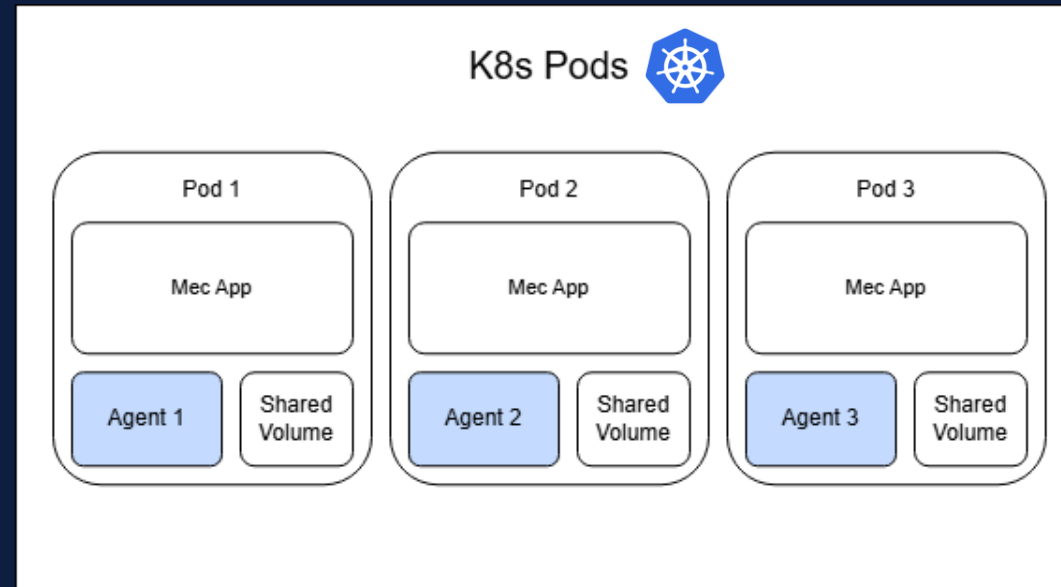
System Architecture



```
NAME
mongo-0
mongo-express-57845fb795-wx1z9
cfs-portal-7b5c8ffc5-sv56f
oss-df7f4c66b-l6js9
meao-fb6d9fcd9-6hmkc
```



Deployment diagram



- Each MEC application is deployed in a pod.
- Each pod's agent reports its container's state (base OS, installed packages, process logs) to SIEM



Dynamic Goals

- Full system analysis
- External threat protection
- Optimal log collection
- Incident response with dedicated software
- Network slicing
- Cloud analysis

